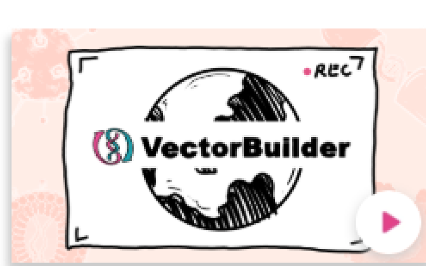
How to Use
Vector Design Studio



About VectorBuilder

Codon Table

Codon tables are the Rosetta Stone of molecular biology; they are a fundamental tool in enabling you to decipher the genetic code. The table below acts as a reference for how nucleotides translate into codons as well as unique species-specific preferences for individual codons. Codon optimization can be used on large-scale sequences, while this table can be used to analyze individual codons. You can clone codon-optimized sequences easily into vectors and order downstream services by clicking here.

Click here to design and order a vector with your chosen sequence in just a few clicks.
Receive plasmid preparations of your sequence at a variety of scales with fast turnaround.
Order high-titer packaging of your vector in ready-to-use virus.

Codon Table | Crash Course | Tips

Select organism: *
Human (Homo sapiens)

Table type: *
○ Intuitional ◉ Proportional

1st Base	2nd Base	Codon	Amino Acid	Frequency	Frequency Graph
T	T	TTT	Phe (F)	0.46	
		TTC	Phe (F)	0.54	
		TTA	Leu (L)	0.08	
		TTG	Leu (L)	0.13	
C		CTT	Leu (L)	0.13	
		CTC	Leu (L)	0.20	
		CTA	Leu (L)	0.07	
		CTG	Leu (L)	0.40	
A		ATT	Ile (I)	0.36	
		ATC	Ile (I)	0.47	
		ATA	Ile (I)	0.17	
		ATG	Met (M)	1.00	
G		GTT	Val (V)	0.18	
		GTC	Val (V)	0.24	
		GTA	Val (V)	0.12	
		GTG	Val (V)	0.46	
T	C	TCT	Ser (S)	0.19	
		TCC	Ser (S)	0.22	
		TCA	Ser (S)	0.15	
		TCG	Ser (S)	0.05	
C		CCT	Pro (P)	0.29	
		CCC	Pro (P)	0.32	
		CCA	Pro (P)	0.28	
		CCG	Pro (P)	0.11	
A		ACT	Thr (T)	0.25	
		ACC	Thr (T)	0.36	
		ACA	Thr (T)	0.28	
		ACG	Thr (T)	0.11	
G		GCT	Ala (A)	0.27	
		GCC	Ala (A)	0.40	
		GCA	Ala (A)	0.23	
		GCG	Ala (A)	0.11	
T	A	TAT	Tyr (Y)	0.44	
		TAC	Tyr (Y)	0.56	
		TAA	Ter (*)	0.30	
		TAG	Ter (*)	0.24	
C		CAT	His (H)	0.42	
		CAC	His (H)	0.58	
		CAA	Gln (Q)	0.27	
		CAG	Gln (Q)	0.73	
A		AAT	Asn (N)	0.47	
		AAC	Asn (N)	0.53	
		AAA	Lys (K)	0.43	
		AAG	Lys (K)	0.57	
G		GAT	Asp (D)	0.46	
		GAC	Asp (D)	0.54	
		GAA	Glu (E)	0.42	
		GAG	Glu (E)	0.58	
T	G	TGT	Cys (C)	0.46	
		TGC	Cys (C)	0.54	
		TGA	Ter (*)	0.47	
		TGG	Trp (W)	1.00	
C		CGT	Arg (R)	0.08	
		CGC	Arg (R)	0.18	
		CGA	Arg (R)	0.11	
		CGG	Arg (R)	0.20	
A		AGT	Ser (S)	0.15	
		AGC	Ser (S)	0.24	
		AGA	Arg (R)	0.21	
		AGG	Arg (R)	0.21	
G		GGT	Gly (G)	0.16	
		GGC	Gly (G)	0.34	
		GGA	Gly (G)	0.25	
		GGG	Gly (G)	0.25	